

Research article

Derivation of novel metabolic pathway score identifies alanine metabolism as a targetable influencer of TNF-alpha signaling

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ABSTRACT

Background: Better understanding of the interaction between metabolism and immune response will be key to understanding physiology and disease. Tumor Necrosis Factor-alpha (TNF α) has been studied widely. However, despite the extensive knowledge about TNF α , the cytokine appears to induce not only variable, but often contradictory, effects on inflammation and cell proliferation. Despite advancements in the metabolomics field, it is still difficult to analyze the types of multi-dose, multi-time point studies needed for elucidating the varied immunologic responses induced by TNF α .

Results: We studied the dose and time course effects of TNF α on murine fibroblast cultures and further elucidated these connections using selective blockade of the TNF receptors (TNFR1 and TNFR2). To streamline analysis, we developed a method to collate the metabolic pathway output from MetaboAnalyst into a single value for the Index of pathway significance (IPS). Using this metric, we tested dose-, time-, and receptor-dependent effects of TNF α signaling on cell metabolism. Guided by these results, we then demonstrate that alanine supplementation enriched TNFR1-related responses in both cell and mouse models.

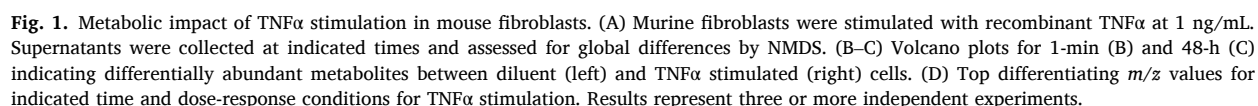
Conclusions: Our results suggest that TNF α , particularly when signaling through TNFR1, may preferentially use alanine metabolism for energy. These results are limited in by cell type used and immune outputs measured. However, we anticipate that our novel method may assist other researchers in identifying metabolic targets that influence their disease or model of interest through simplifying the analysis of multi-condition experiments. Furthermore, our results endorse the consideration of follow up studies in immunometabolism to improve outcomes in TNF-mediated diseases.

1. Introduction

The understanding of the mechanistic connection between metabolism and immune function continues to expand in modern research. One prominent example is the metabolic reprogramming that underlies innate cellular immune memory [1]. Additionally, a recent report demonstrates that interferon gamma (IFN γ) signaling requires the nicotinamide phosphoribosyl transferase-nicotinamide adenine dinucleotide phosphate (NAMPT-NADPH) pathway to activate macrophages in culture [2]. The capacity for metabolomic analysis in immunology has greatly expanded via advances in mass spectrometry including improved proteomics, integration of RNAseq with metabolism, and spatial metabolomics [3–5]. Furthermore, programs, such as MetaboAnalyst, take large data sets on individual metabolisms and allow researchers to organize them into more manageable metabolic pathways [6]. However, evaluation of cytokine responses in immunology often requires experimental consideration of dosage, time course, and co-stimulatory factors. Each of these generate the need for multiple conditions. Despite how well metabolomic analyses can handle

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large numbers of analytes, the analytic approaches are often limited in the number of experimental conditions they can compare. In this report we aimed to achieve two goals: (1) To develop a method to collate the available metabolic pathway data into a single value; and (2) assess actionable metabolic differences in the cellular response to tumor necrosis factor alpha (TNF α) using this method.

TNF α can mediate seemingly opposing responses of inflammation and immunity because of its ability to operate through two different receptors, TNF receptor 1 (TNFR1) and TNFR2. TNFR1 is associated with inflammation and apoptosis [7] while TNFR2 aids in cell activation, migration, and proliferation [8]. However, the standard characterization of TNFR1 as pro-inflammatory and TNFR2 as immune-modulatory can vary by disease model [9,10]. Although of great clinical utility, because all current TNF α -blocking agents preclude the signaling pathways by binding to the TNF ligand through either receptor [9], current therapies leave many unanswered questions about optimizing clinical responses through differential impacts on the TNF receptors. Given that several papers have described the metabolic effects of TNF α on other organ systems [11–14], we hypothesized that the given metabolic conditions of immune cells may contribute to differential signaling between the TNF α receptors.

Herein we describe that collating MetaboAnalyst output data into a single value for the Index of Pathway Significance (IPS) suggests that dose and time dependent effects of TNF α signaling are heavily influenced by alanine metabolism. We further demonstrate that alanine supplementation alters TNF response in both cell and mouse models.

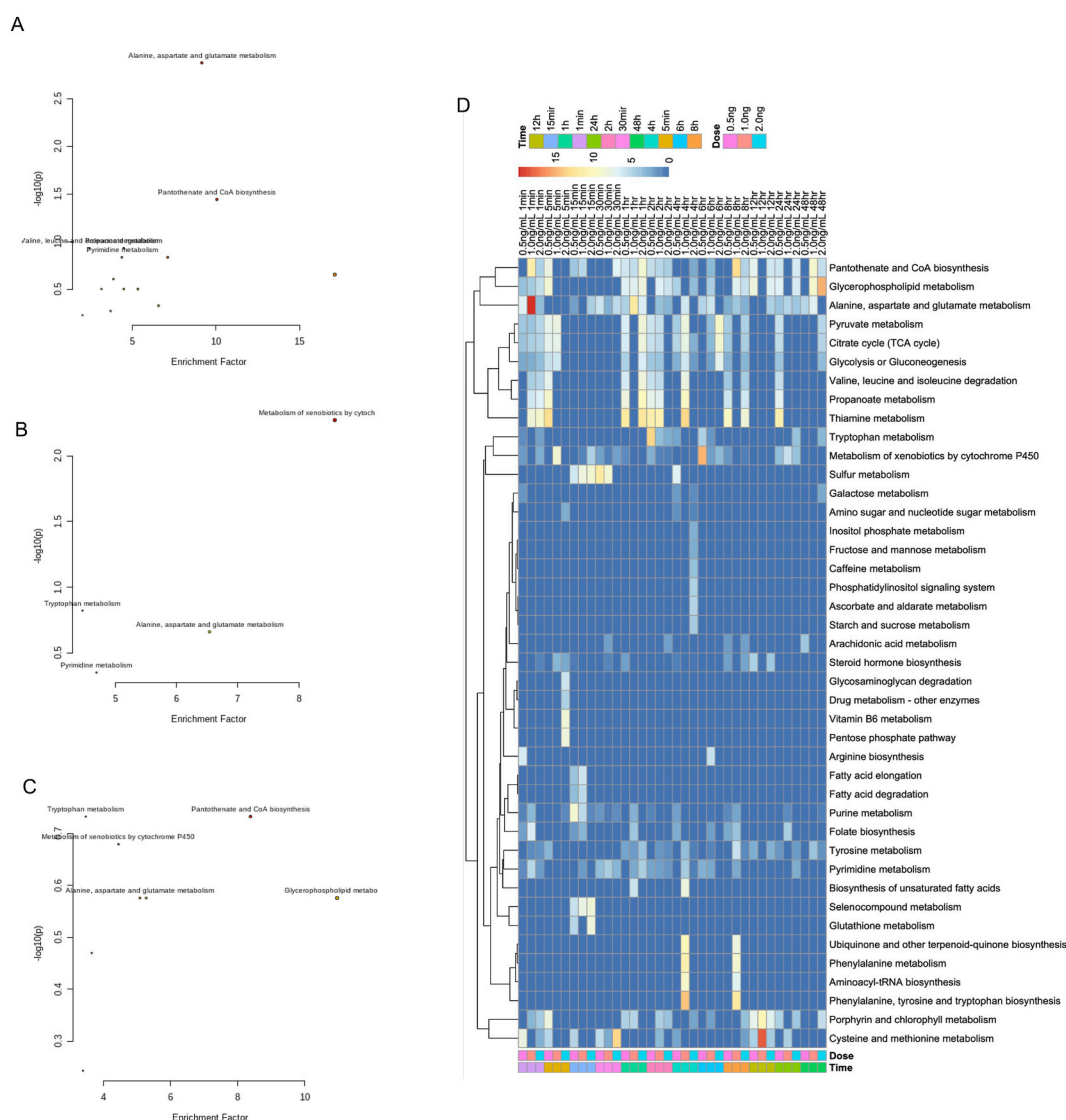


Fig. 2. Pathway analysis of TNF α stimulated fibroblasts implicates alanine metabolism. (A–C) MetaboAnalyst pathway output for 1 ng/mL TNF α for 1 min (A), 0.5 ng/mL at 6 h (B), and 2 ng/mL at 24 h. (D) Summarized IPS values derived from MetaboAnalyst pathway output for all indicated doses and timepoints. Results represent three or more independent experiments.

2. Results

2.1. *TNF α* exposure generates select differences in fibroblast metabolism

Metabolic profiles of *TNF α* -treated 3T3 fibroblasts were compared to those of untreated cells. At each of the 12 time points, the supernatant from each well, along with cell lysates, were collected for metabolomic analysis. The overall similarity could be assessed by non-metric multidimensional scaling (NMDS) plots (Fig. 1a, Supplemental Fig. 1). NMDS is a multivariate statistical technique used

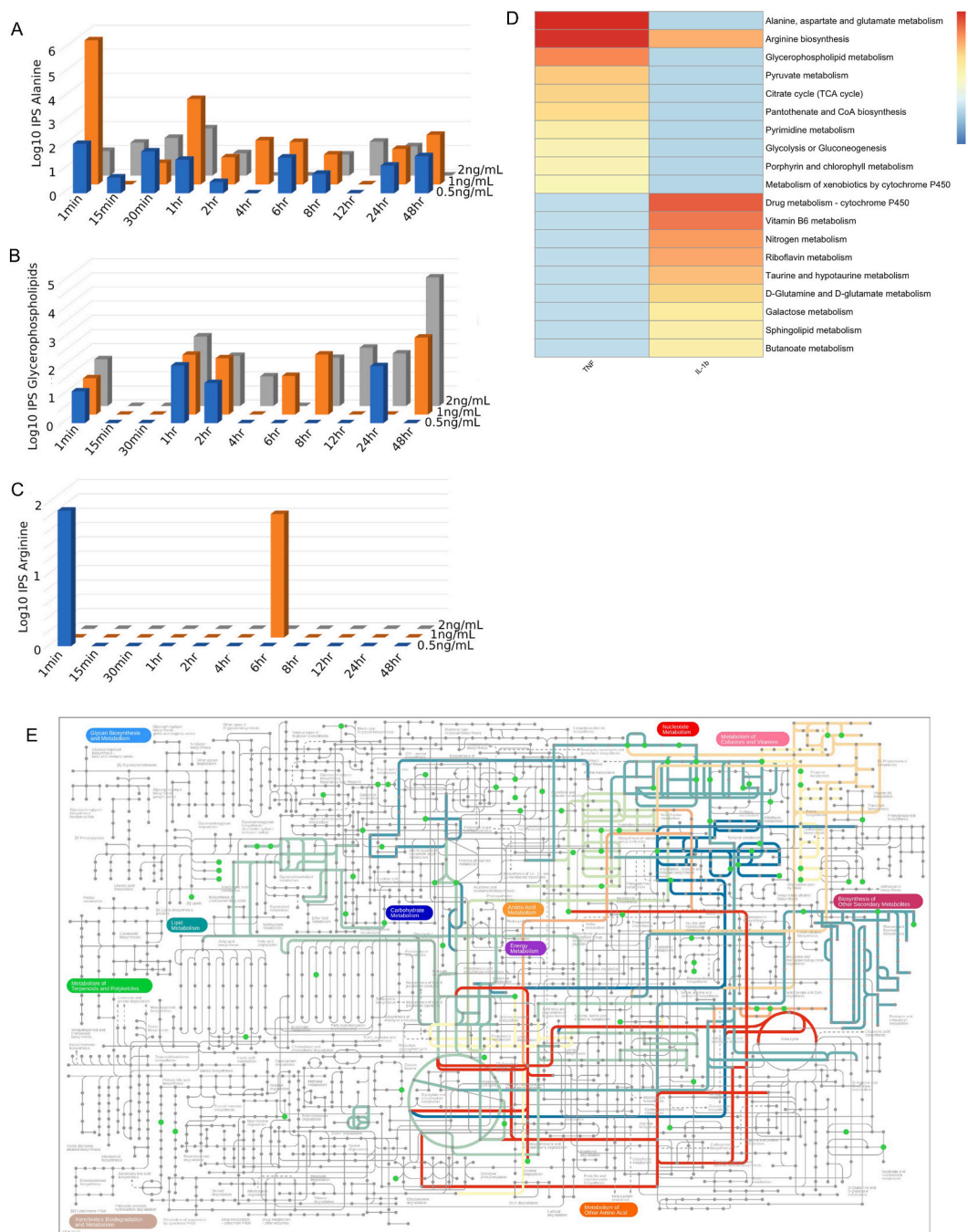


Fig. 3. IPS values allow for multi-condition comparisons. (A–C) IPS values for matrix of time and dose differences in *TNF α* stimulated fibroblasts for alanine metabolism (A), glycerophospholipids (B), and arginine metabolism (C). (D) IPS values for fibroblasts stimulated with 1 ng/mL of either *TNF α* or IL-1 β . (E) IPS values super imposed onto KEGG metabolic pathway summary. Results represent three or more independent experiments.

for exploring and visualizing similarities or dissimilarities between objects or samples in a dataset. Unadjusted significance indicates differences in global metabolism at the 1-min, 1-h, and 48-h timepoints; however, adjusted values suggests no global differences arise with $\text{TNF}\alpha$ treatment of the cells. However, the intensity values for numerous individual peaks are significantly different at several timepoints (Fig. 1b–c, Supplemental Fig. 2). No single metabolite is significant across all conditions.

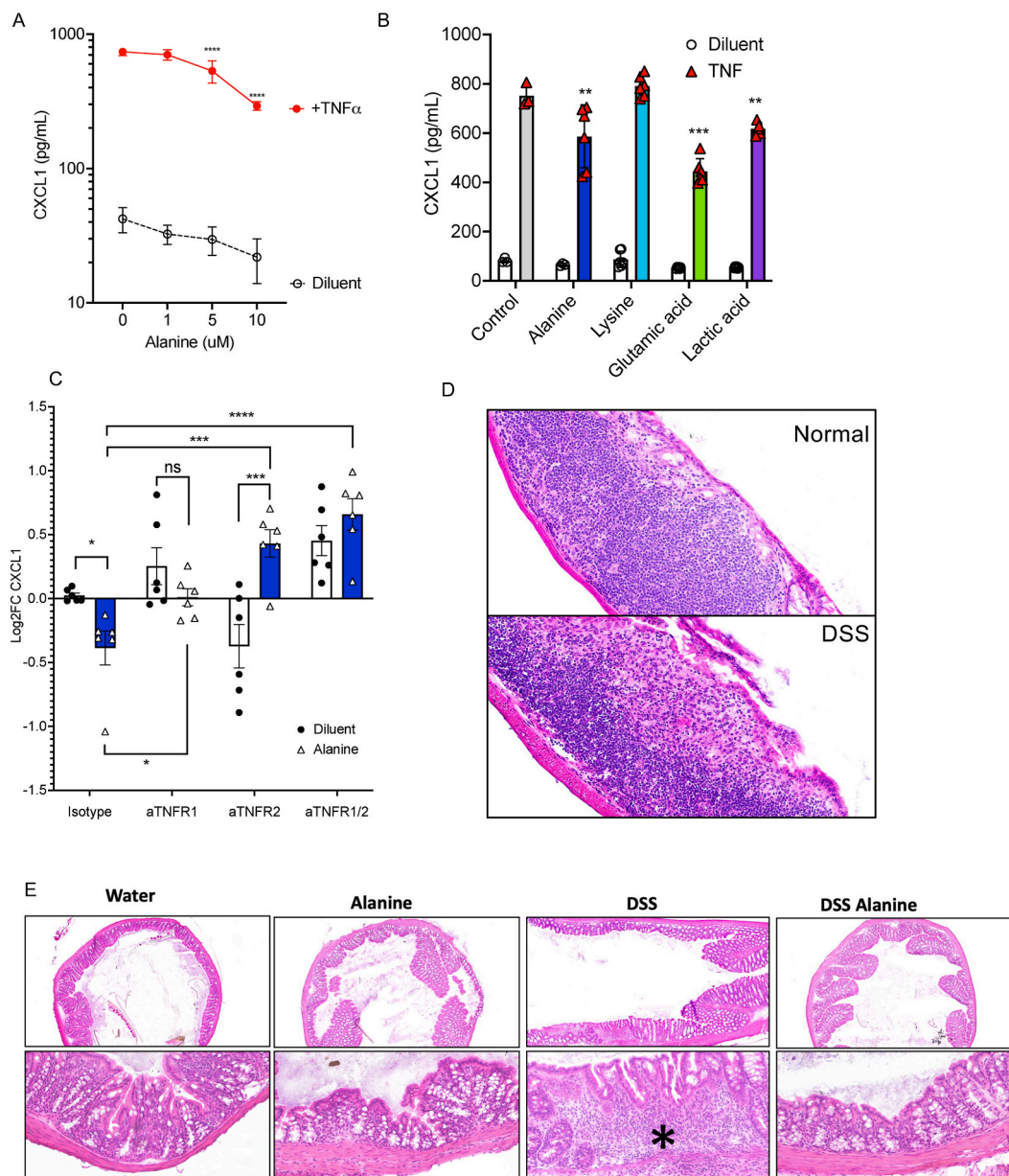


Fig. 4. Alanine exposure modulates models of TNF-induced immunity. (A) CXCL1 supernatant concentrations for mouse fibroblasts stimulated with $\text{TNF}\alpha$ (1 ng/mL) and indicated concentrations of alanine. (B) CXCL1 supernatant concentrations for mouse fibroblasts stimulated with $\text{TNF}\alpha$ (1 ng/mL) and 5 μM doses of indicated substance. (C) Log2 fold change (FC) for CXCL1 supernatant concentrations for mouse fibroblasts stimulated with $\text{TNF}\alpha$ (1 ng/mL), 5 μM alanine, with either isotype control or antibody blockade for TNF receptor 1 (TNFR1) and/or TNFR2. (D–E) Mice (N = 5 mice per group) were given water supplemented with 5 μM alanine, 5 % DSS, or 5 μM alanine + DSS. Colons were collected after 10–14 days of exposure and assessed for histologic evidence of inflammation. Images in D compare normal colonic gut associated lymphoid tissue (GALT) to the neutrophilic inflammation with mucosal ulceration in the GALT of a DSS mouse. In D, the segmental acute inflammation (*) in the lamina widely separating crypts (i.e. colitis) in the DSS-only mouse. Colons from mice given water, alanine, or DSS plus alanine are essentially normal. Results represent four (A–C) or two (D–E) independent experiments and are depicted as mean $\pm$ SEM. *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, **** $p < 0.0001$; ns = not significant as determined by Student T test (A) or ANOVA (B–C).

2.2. Collating MetaboAnalyst output into a single value enhances multi-condition comparisons

MetaboAnalyst collates individual peak identifications into metabolic pathways defined by databases such as the Kyoto Encyclopedia of Genes and Genomes (KEGG). Although the visual (Fig. 2a, Supplemental Fig. 3) and numerical outputs reveal the metabolic pathways that demonstrate pathways with the greatest differences between two treatment conditions (Supplemental Fig. 3).

To address some of the limitations in pathway visualization, we developed a novel formula, the Index of Pathway Significance (IPS; see methods for details on derivation) that assigns a single, easily compared value to each pathway. IPS values are compared to each of the three TNF-treated conditions with the control group at each of the respective time points. This reveals that the largest differences are in alanine, aspartate, and glutamate metabolism (Fig. 2b). Use of IPS also facilitates comparisons within time points across various doses of TNF α stimulation (Fig. 3a–c) to establish pathways with consistent signatures rather than those with likely spurious values (Fig. 3c). Furthermore, interleukin (IL)-1 β -treated fibroblasts demonstrate that the metabolic signature of TNF α activation is not shared with IL1 β stimulated cells (Fig. 3d). The IPS values can also be superimposed upon the KEGG summary pathways to show the intensity that each pathway is altered by TNF α exposure (Fig. 3e; Supplemental Fig. 4).

2.3. Alanine impact on CXCL1 is TNFR1 dependent

Alanine was supplemented into cell media to investigate the effects of TNF α under alanine-enriched conditions. CXCL1 is a chemottractant for immune cells, including neutrophils, and its release is stimulated by TNF α [15]. Because of its known involvement in cell migration [16], as well as inflammation [17], this chemokine was used as a representative measurement of functioning immune response. C-X-C motif ligand 1 (CXCL1) concentrations are higher in the supernatant of cells challenged with TNF α ; alanine supplementation results in significantly less CXCL1 release compared to diluent in a dose dependent manner (Fig. 4a). Neither lysine, glutamic acid, nor lactic acid significantly impact CXCL1 production in TNF-naïve cells (Fig. 4b). However, glutamic and lactic acid inhibit CXCL1 from TNF-treated fibroblasts (Fig. 4b), suggesting a partial role for pH in the effects of alanine.

As before, when both TNF receptors are available, alanine inhibits CXCL1 production. However, the significance of this effect is ablated by blockade of TNFR1 (Fig. 4c). While alanine continues to alter CXCL1 production under blockade of TNFR2, the impact appears paradoxical (Fig. 4c). Effects are not significant when the cells are challenged with either glutamic acid or lysine (Supplemental Figs. 5a–b). Supplementation with lactic acid significantly enhances CXCL1 release when TNFR1 is blocked (Supplemental Fig. 5c).

2.4. Alanine supplementation normalizes outcomes in DSS colitis

Given prior reports of the TNFR1 dependence of the murine DSS colitis model [10], we supplemented DSS drinking water with 5 nM alanine. 40 % of mice in the DSS treated group demonstrate either focal extensive neutrophilic inflammation effacing the gut associated lymphoid tissue (GALT) or segmental neutrophilic infiltration in the lamina propria that separates and replaces crypts and extends into the submucosa (Fig. 4d–e). However, these effects were ablated by alanine supplementation (Fig. 4d). Alanine treatment alone did not impact weight change or histology.

3. Discussion

Analysis tools such as those provided by MetaboAnalyst have greatly improved metabolomic workflows by summarizing complex data sets of individual metabolites into pathways that may be experimentally investigated for therapeutic potential. In this work, we sought to expand on the ability to perform pathway analysis across numerous conditions of dose, time, and cofactor exposure. To our knowledge, the effects of TNF α on metabolism have not been thoroughly studied at the cellular level in immune cells, despite descriptions of metabolic impacts in non-immunologic tissue [11–14].

TNF α is a cytokine primarily produced by activated macrophages or T lymphocytes [7], but also expressed by a variety of other cell lines [18]. When existing as a transmembrane protein, it has a molecular mass of 26 kDa, however, can be secreted as a 17 kDa protein under certain conditions with the help of TNF-alpha-activating converting enzyme (TACE) [19]. While TNF α levels are typically undetectable in healthy individuals [7], abnormal concentrations in the serum, tissue, or stool are associated with a wide array of diseases, including asthma [20], HIV [21], Crohn's disease [22], diabetes [23], and psoriasis [24], among others. Furthermore, serum levels frequently correlate with the severity of the disease [25].

Our investigation identifies alanine, aspartate, and glutamate metabolism as being the most significantly different pathway between TNF-treated fibroblasts. Supplementation with alanine alters the CXCL1 production of these cells in a TNFR-sensitive manner and could be used to improve outcomes in the TNFR-dependent model of DSS colitis [10]. This finding provides additional detail about the underlying cellular activity that goes beyond the heavily studied bioactivities associated with each of the two TNF receptors.

Along with differences in structure and signaling pathways [26,27], TNFR1 and TNFR2 can differ with respect to percentage of cells expressing each type of receptor, as well as the number of receptors per cell. For example, when looking specifically at peripheral immune cells, both measurements have been found to be higher for TNFR2 compared to TNFR1 [28]. That said, while TNFR1 is nearly omnipresent [29], TNFR2 is limited to certain cell types, including myeloid cells, regulatory T-cells, glial cells, and some endothelial cells, along with being induced in epithelial cells and fibroblasts [30]. TNFR1 and TNFR2 differ further when discussing the TNF α molecules that they can bind and their preferences for each. While transmembrane TNF α (tmTNF α) can efficiently activate TNFR1 and TNFR2 signaling pathways, soluble TNF (sTNF α) only activates TNFR1 to a high degree [31]. This is due to sTNF α 's low attraction to

TNFR2, and the dissociation that is still possible after initial binding [32].

Our work suggest that the metabolic state of fibroblasts may exert additional influence over TNF α signaling in general as well as specific responses to the differential signaling through the TNF receptors. In the light of the literature indicating that TNFR1 ablation worsens outcomes in DSS colitis [10], our cell and mouse model results suggest that alanine supplementation preferentially enhances signaling through TNFR1 and suggests opposing impacts on CXCL1 by TNFR1 and TNFR2. Furthermore, the results indicating that alanine's impact on CXCL1 is TNFR1-dependent suggest a bi-directional crosstalk between immunity and metabolism. Alanine's role in the alanine-cycle may lend to future experiments using inhibitors of glutamic-pyruvic transaminase (GPT) to assess their impact on TNF α signaling. In addition, in vitro systems capable of discerning TNFR1 versus TNFR2 activation in real time would greatly aid the field.

However, our work is limited in several ways. First, our findings are restricted to fibroblast cultures and murine models of general TNF response. These findings would greatly benefit from mouse model validation in additional disease models that are known to be influenced by TNF such as tuberculosis or rheumatic ailments [9] as well as challenging other cell types that are TNF α modulated. Second, we limited our functional output to CXCL1 production for simplicity, but the potential effects of TNF α go far beyond this single chemokine. Third, while alanine supplementation had differing impacts than glutamic acid or lactic acid, the possible contribution of pH cannot be excluded; however, separating the impacts of select amino acids without chemically altering them via buffers is not likely to be a viable approach. Finally, while our IPS formula was derived systematically and validated previously [33], alternative equations could lead to different results. However, while we cannot present this work as a definitive analysis of the metabolic impact of TNF α , we were able to use our novel metabolic pathway metric to identified alanine as a modulator of TNF α response in both cell and mouse models.

Overall, our results offer a novel addition to current metabolomic workflows that may allow researchers to assess large numbers of conditions more readily for metabolic pathways with the potential for therapeutic manipulation. Ultimately, we hope that the integration of the IPS into metabolic workflows may further enhance the identification of targetable metabolic pathways that may improve immunologic outcomes.

4. Materials and methods

4.1. Cell culture

NIH/3T3 Mouse Fibroblasts were previously purchased from ATCC. A frozen cryovial was thawed and cultured in DMEM (gibco; #11965092) containing 10 % FBS and 100 μ g/mL penicillin/streptomycin glutamine (gibco; #15140122). After nearing full confluence in a T75 flask, cells were trypsinized with 0.05 % trypsin-EDTA (gibco; #25300054) and seeded on 24 well plates (corning; #3524) at 70,000 cells per well concentration, one for each time point of interest. Plates were then incubated for a minimum of 24 h, or until 75 % confluence was achieved, and each treatment condition was plated in triplicates. After 24 h, cells were treated with different conditions and cell supernatant, as well as cell lysates, were collected and stored at -80°C until further use.

Cells were exposed to any added organic compounds and/or antibodies 6 h before start of the experiment ($t = -6$), and 550 μ L of fresh DMEM media containing the same supplements, as well as TNF α (#300-01A) from Peprotech when appropriate, was added at $t = 0$ at 0.5 ng/mL to 2 ng/mL as indicated. Plates were then returned to the incubator until their specific time point.

4.2. In vitro sample collection

At each time point, the specific plate was taken from the incubator and cell supernatant was transferred from each well to a different Eppendorf tube. In the initial exploratory trial, 250 μ L of pure methanol at room temperature was then added to each well for cell lysis. This volume was later adjusted to 500 μ L. Cells remained in the methanol for 20 min, and occasional mechanical irritation of the plate was used to promote the rupturing. Once complete, the contents of each well were removed and added to the same Eppendorf tube as the supernatant in the case of the exploratory trial, and different tubes in subsequent trials. All tubes were then frozen at -80°C Fahrenheit.

4.3. Mass spectrometry

To assist in mass spectrometry data collection, a matrix consisting of 90 % MeOH, 20 mg/mL 2,5-dihydroxybenzoic acid matrix (Sigma Aldrich; 85,707), and 0.01 % TFA (Sigma; T6508) was prepared. Collected samples were thawed and a 3:2:1 ratio of matrix: supernatant:lysate was prepared for each individual sample. 2 μ L from each prepared solution was spotted onto a metal spot plate and run through MALDI timsTOF Fleximager (Bruker Daltonik, Bremen, Germany). Specifically, the machine was operating in TIMS qTOF positive ion mode from m/z 200 to 2000 with a laser intensity of 75 %. Additional settings were as previously described [5].

4.4. Index of Pathway Significance

The Index of Pathway Significance (IPS) is an in-house formula derived from five metrics provided by MetaboAnalyst: total number of metabolites in a metabolic pathway, the total number of metabolite hits picked up from a sample, the number of significant metabolite hits in a pathway, the expected number of total metabolite hits in a sample, and FET, the p-value associated with a sample. When considered together, they produce a single, easily comparable value that is associated with differences between two treatments

with respect to a certain metabolic pathway. A higher IPS values indicates a greater difference and can be used to rank either the metabolic pathways that are shared between two treatments, or the treatments that all exhibit that particular pathway. Assigning a single value to each pathway also allows many comparisons to be condensed and visualized in figures such as heat maps and metabolic pathway maps.

In the formula, the placement of each of the five metrics was logically determined based on how its manipulation would affect the final IPS value. The numerator consists of the weighted sum of significant and non-significant metabolites that were found in the samples of interest and divides it by the total number of metabolites in that same pathway. While all metabolites that were found to be different between the samples of interest are important, those whose differences were statistically significant were deemed more so, and consequently, the number of significant metabolites was squared. The total number of metabolites in a given pathway was placed in the denominator of the ratio, as the number of identified metabolites is expected to increase as the number of identifiable metabolites increases. In other words, with all other metrics being equal, identifying five metabolites in a pathway with ten total metabolites is more significant than identifying five metabolites in a pathway with fifty total metabolites. This ratio is then divided by the number of metabolites from that pathway that MetaboAnalyst expected to find in the samples, with a lower value further underscoring the significance of any and all identified metabolites. With FET, or p-value, being perhaps the metric most indicative of differences between two samples, it is squared and positioned in the denominator to increase the final IPS score exponentially as it approaches or passes statistical significance. Finally, to avoid calculating the same IPS value when the number of significant metabolites is zero and one, respectively, one is added to the final quotient.

To validate the placement of each of the five metrics, as well as the degree to which they affected the final IPS value, arbitrary data was created, and each metric was manipulated individually.

4.5. Cytokine assay

After experiments, cultured cell supernatant was stored at -80°C before being used to measure cytokine level. We have used the Bio-Plex Pro mouse Th17 cytokine assay kit from Bio-Rad (12010828) and protocol was followed according to manufacturer instructions. 50 μl of diluted beads were added to each well of the assay plate. Followed by washing the plate with 100 μl of Bio-Plex wash buffer. 50 μl of samples, standard, and a blank were added to respective wells. The plate was then incubated in the dark for 1 h at room temperature with vigorous shaking before being washed 3 times with a wash buffer and the addition of 25 μl of detection antibodies in each well. This was followed by incubating the plate in the dark for 30 min with vigorous shaking, and the plate was washed 3 times with a wash buffer. Then, 50 μl 1 $\times$ streptavidin-PE (SA-PE) was added in each well before the plate was incubated in the dark for 10 min with vigorous shaking. Finally, we washed the plate three times with 100 μl wash buffer, resuspend beads in 125 μl assay buffer, covered it, and placed it in the shaker at 850 rpm for 30 s. Reading samples were taken with the Bio-Plex 200 system from Bio-Rad.

4.6. Mice

All mice were purchased from the Jackson laboratory. C57BL/6 mice aged 6–10 weeks were used for experiments with age and sex matched within each experiments. 5 % DSS (Sigma) was dissolved into the mouse drinking water. Mice ($N = 5$ per group) were allowed to drink ad libitum for 10–14 days prior to collection of histology tissue. Tissues were mounted and stained with H&E by Histoserv Inc (Germantown, MD). Colon sections were examined by light microscopy using an Olympus BX51 microscope and photomicrographs were taken using an Olympus DP28 camera. Mice experiments were performed under the approval and supervision of the NIAID Animal Care Protocols.

4.7. Statistics

All MALDI imaging data were visualized using SciLS Lab Version 2021 (Bruker Daltonics). Statistical analyses were conducted in R, including hierarchical clustering. Intensity data was pareto scaled and log transformed. ANOVA was used to identify group differences when >2 groups were compared, Student T test was used when only 2 groups were compared. P-values were adjusted using the Benjamini and Hochberg method.

Ethics statement

This study was reviewed and approved by the Animal Study Protocol committee of the National Institute of Allergy and Infectious Diseases, with the approval number LCIM8E.

Data availability

All data are included in the article of supplemental materials. Raw metabolomics files can be shared upon reasonable request.

CRedit authorship contribution statement

Brandon N. D'Souza: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Manoj Yadav:** Writing – review & editing, Writing – original draft,

Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Prem Prashant Chaudhary:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis. **Grace Ratley:** Writing – review & editing, Visualization, Validation, Methodology. **Max Yang Lu:** Writing – review & editing, Visualization, Validation, Investigation, Data curation. **Derron A. Alves:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis. **Ian A. Myles:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.heliyon.2024.e33502>.

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